

SHUBHAM RANA

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ACADEMIC DETAILS

Education	Institute	Year	CGPA/%
Bachelor of Technology (CSE)	Graphic Era University, Dehradun	2023 - Present	8.86/10
Senior Secondary Education (Class XII), CBSE	Kendriya Vidyalaya, F.R.I	2023	94.2%
Secondary Education (Class X), CBSE	Kendriya Vidyalaya, F.R.I	2021	97.8%

SKILLS

Programming Languages	Web Development	Databases	Soft-Skills	CS Fundamentals	Tools & OS
C++	HTML5	PostgreSQL	Problem Solving	Computer Networks	VS Code
C	CSS3	MongoDB	Teamwork	Data Structures and Algorithms	Git (GitHub)
Python	JavaScript	MySQL	Leadership	Object Oriented Programming	SQL
Java	Tailwind CSS			Database Management System	Linux
	React.js			Operating System	Windows
	Node.js				
	Express.js				
	Next.js				

PROJECTS

- **ReClaim** (A database driven lost and found management system.) (March 2025 - July 2025)
<https://github.com/Shubham-Rana-cse/ReClaim--Database-Driven-Lost-and-Found-Management-System>
 - A web-based application to digitize the conventional lost & found process, enhancing efficiency and user accessibility. Enabling image upload functionality with a 16 MB size limit, support for 4 validated image formats.
 - Built with a responsive frontend with 11 HTML templates, integrated with a backend with 14 flask endpoints ensuring role-based access control, authentication, and CRUD operations upon 6 PostgreSQL tables.
 - Tech-stack used: HTML | CSS (Tailwind CSS) | JavaScript | Python | PostgreSQL
- **Route Optimizer** (using an algorithm for solving Travelling Salesman Problem) (September 2025-December 2025)
<https://your-route-optimizer.netlify.app/>
 - Developed a React-based route optimization tool with 7 modular UI components, with interactive Leaflet/OpenStreetMap interface. It supports up to 5 location suggestions per search query and 300 ms debounced search for faster input handling.
 - Implemented an $O(n^2)$ distance matrix using OSRM road-distance APIs with Haversine fallback and then applying nearest-neighbor Travelling Salesman heuristic to generate optimized route.
 - Tech-stack used: HTML | CSS (Tailwind CSS) | JavaScript | React | Leaflet-'Open-Street-Map'
- **"Melodify"** (A web music player heavily inspired from "Spotify- Web Player") (November 2024)
<https://melodify-app.netlify.app/>
 - Single-page HTML-CSS-JS based music player; making 4 fetch-based operations to load songs/albums dynamically from album & song directories; player providing 5 playback controls: shuffle, previous, play/pause, next, and repeat.
 - Tech-stack used: HTML | CSS | JavaScript
- **"PassMan"** (MERN based password manager application) (February 2026)
<https://github.com/Shubham-Rana-cse/PassMan-MERN>
 - React-based app with 4 UI components, a 3-field credential form, animated icons, a responsive table-based vault for managing saved passwords, clipboard copy, password visibility toggle, and toast-based user feedback across 5 entry actions to improve usability.
 - Express + MongoDB backend with 3 REST-style endpoints for reading, saving, and deleting credentials from a single passwords collection.
 - Tech-stack used: React| JavaScript | Tailwind CSS | Express.js | Node.js | MongoDB

ACHIEVEMENTS

- **Academic achievements**
 - Qualified All India National Defense Academy Examination, 2023 (conducted by UPSC).
 - Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate.
- **Other achievements**
 - Bronze medalist in 4*100 m relay, in annual athletics meet, organized by Graphic Era University.